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Original Research

Uncovering hidden subtypes in dementia: An unsupervised machine learning approach to dementia diagnosis and personalization of care

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ABSTRACT

Objective: Dementia represents a growing public health challenge, affecting an increasing number of individuals. It encompasses a broad spectrum of cognitive impairments, ranging from mild to severe stages, each of which demands varying levels of care. Current diagnostic approaches often treat dementia as a uniform condition, potentially overlooking clinically significant subtypes, which limits the effectiveness of treatment and care strategies. This study seeks to address the limitations of traditional diagnostic methods by applying unsupervised machine learning techniques to a large, multi-modal dataset of dementia patients (encompassing multiple data sources including clinical, demographic, gene expression and protein concentrations), with the aim of identifying distinct subtypes within the population. The primary focus is on differentiating between mild and severe stages of dementia to improve diagnostic accuracy and personalize treatment plans.
Methods: The dataset analyzed included 911 individuals, described by 157 multi-modal characteristics, encompassing clinical, genomic, proteomic and sociodemographic features. After handling missing data, the dataset was reduced to 561 rows and 135 columns. Various dimensionality reduction techniques were applied to improve the feature-to-patient ratio, and unsupervised clustering methods were employed to identify potential subtypes. The major novelty in our methodology regards the combination of different techniques, bridging high-dimensional statistical inference, multi-modal dimensionality reduction and clustering analysis, to appropriately model the multi-modal nature of the data and ensure clinical relevance.
Results: The analysis revealed distinct clusters within the dementia population, each characterized by specific clinical and demographic profiles. These profiles included variations in biomarkers, cognitive scores, and disability levels. The findings suggest the presence of previously unrecognized subgroups, distinguished by their genomic, proteomic, and clinical characteristics.
Conclusion: This study demonstrates that unsupervised machine learning can effectively identify clinically relevant subtypes of dementia, with important implications for diagnosis and personalized treatment. Further research is required to validate these findings and investigate their potential to improve patient outcomes.

1. Introduction

Dementia represents a significant public health challenge [1], affecting an increasing number of individuals globally, largely due to the ongoing demographic transition, which is leading to a substantial rise in the elderly population [2]. The spectrum of cognitive decline ranges from mild cognitive impairment, where patients maintain some level of independence, to severe dementia, where extensive care and support are required. The most common form of dementia is Alzheimer's disease (AD), followed by vascular dementia (VaD). While these two conditions are traditionally regarded as distinct from a clinical standpoint, recent evidence suggests that cerebral vascular damage plays a key role in cognitive decline [3], and vascular alterations are observed across all major forms of dementia, including neurodegenerative types [4].

In light of these findings, a new continuum model of dementia has been proposed, suggesting that damage to the cerebral vasculature

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may precede the neurodegenerative changes typical of AD [5]. According to the American Heart Association/American Stroke Association (AHA/ASA), vascular and neurodegenerative disorders frequently overlap, often coexisting in the same individual and contributing to what is known as mixed dementia [4]. Mixed dementia is particularly prevalent among older adults [6–10], complicating diagnosis, prognosis, and treatment.

Understanding the underlying patterns and subgroups within the dementia population can provide critical insights for the development of targeted interventions and personalized care strategies. However, current diagnostic criteria often treat dementia as a uniform condition, particularly when differentiating between AD and VaD [11]. However, this approach may overlook subtle, yet clinically significant, variations within these categories, potentially resulting in suboptimal management strategies. Identifying clusters or subtypes within the dementia population could lead to a more nuanced understanding of the disease and support the development of personalized treatment plans.

The current body of knowledge on dementia encompasses extensive research on its clinical presentation, progression, and associated risk factors. However, much of this research relies on conventional statistical methods, which may fail to capture complex, non-linear relationships within the data [12,13]. While some studies have attempted to identify subgroups of dementia patients, these efforts have often been limited by small sample sizes, methodological constraints, or a focus on specific biomarkers, rather than taking a holistic view of patient characteristics. To date, comprehensive applications of unsupervised machine learning techniques to large-scale clinical data from dementia patients remain scarce [12,14]. Existing studies that incorporate machine learning typically emphasize supervised learning approaches [15], which aim to predict outcomes based on predefined labels. In contrast, unsupervised learning – unrestricted by predefined labels – has the potential to uncover previously unrecognized patterns or groups within the data. Despite its potential, this approach remains under-explored in dementia research, as evidenced by a literature search we conducted through the Scopus database[1] [16–24]. Grossi et al. [20] applied self-organizing maps (a clustering algorithm based on neural networks) to stratify patients affected by Alzheimer's disease, using multi-modal data which encompassed clinical, demographic, biochemical and neuropsychological data (not genomic nor proteomic). Duarte et al. [17] employed an ad-hoc feature reduction method for positron emission tomography neuroimaging data, aimed at improving clustering analysis to diagnose Alzheimer's disease. Durieux et al. [18] applied independent component analysis to fMRI (functional magnetic resonance imaging) data, to facilitate the application of clustering algorithms aimed at stratifying patients affected by either Alzheimer's disease or behavioral-variant frontotemporal dementia. Gan et al. [19] applied a graph-fusion approach to perform dimensionality reduction in fMRI images with the aim of contrasting the curse of dimensionality in the diagnosis of frontotemporal dementia. Aow et al. [16] applied principal component analysis to study conversion from mild cognitive impairment to Alzheimer's disease using MRI data. Simon et al. [23] applied graph-theoretic methods to stratify Parkinson's disease patients in terms of severity classes, using magnetoencephalography. Wang et al. [24] employed a parcellation-based clustering algorithm to stratify mild cognitive impairment and Alzheimer's disease patients based on fMRI data. Huang et al. [21] applied minorization–maximization algorithms for dimensionality reduction in the analysis of neuroimaging data, to study progression from mild cognitive impairment to Alzheimer's disease. Katabathula et al. [22] applied mixed data clustering to stratify patients affected by mild cognitive impairment, combining demographic, neuropsychological and MRI volumetric data, comorbidities and a limited panel of biochemical values.

Table 1
Statement of significance.

Problem	Traditional diagnostic methods for dementia often fail to account for the multifaceted nature of the disease.
What is already known	Dementia encompasses a broad spectrum of cognitive impairments, ranging from mild to severe stages, each requiring different levels of care. However, machine learning studies often treat dementia as a uniform condition, potentially overlooking clinically significant subtypes. Additionally, there is a gap in understanding how various modalities of patient data interact to define distinct subgroups within the dementia population.
What this paper adds	This study aims to overcome the limitations of traditional diagnostic methods by utilizing unsupervised machine learning techniques on a large, multi-modal dataset of dementia patients. The dataset integrates diverse data sources, including clinical, demographic, gene expression, and protein concentration information. The primary objective is to identify distinct subtypes within the dementia population, with a particular focus on differentiating between mild and severe stages to enhance diagnostic accuracy and support personalized treatment plans.

Current limitations stem from reliance on traditional analytical methods, which may not fully utilize the depth and complexity of available data. In particular, there is a gap in understanding how various modalities of patient data and different data sources, such as cognitive scores, imaging results, and genetic markers, interact to define distinct subgroups within the dementia population (as an example, only 2 of the surveyed studies employed more than one modality). Another challenge is the need for methods that can effectively manage the high dimensionality and complexity of clinical data, where relationships between variables can be multifaceted. Finally, previous research has predominantly focused on specific forms of dementia, primarily Alzheimer's disease (7 out of 9 of the surveyed studies focused primarily on Alzheimer's disease) and, to a lesser extent, mild cognitive impairment. This approach has largely overlooked the clinical heterogeneity of dementia, particularly in differentiating between mild and severe stages of the disease or among various forms of dementia in real-world settings.

In this study, we address these limitations by applying an integrated unsupervised machine learning approach, combining dimensionality reduction techniques for managing multiple data modalities with clustering approaches and high-dimensional statistical inference, to a large dataset of patients with mild and severe dementia. Our findings reveal distinct clusters within the dementia population, characterized by specific clinical and demographic profiles. These clusters may represent previously unrecognized subtypes of dementia, with significant implications for diagnosis, prognosis, and treatment. By addressing gaps in current knowledge, our research offers a more nuanced understanding of dementia and paves the way for more personalized and effective care strategies (see Table 1).

[1] The query used for the literature search is TITLE-ABS-KEY ((“clustering” OR “unsupervised learning” OR “cluster analysis”) AND (“dementia” OR “mixed dementia”) AND (“multi-modal” OR “high-dimensional”)) AND (LIMIT-TO (DOCTYPE, “ar”) OR LIMIT-TO (DOCTYPE, “cp”)) AND (LIMIT-TO (LANGUAGE, “English”)). The query returned 16 articles, six of which was excluded for not being pertinent (one of them studied delirium rather than dementia, four only applied supervised learning techniques, two were literature reviews focused on supervised learning).

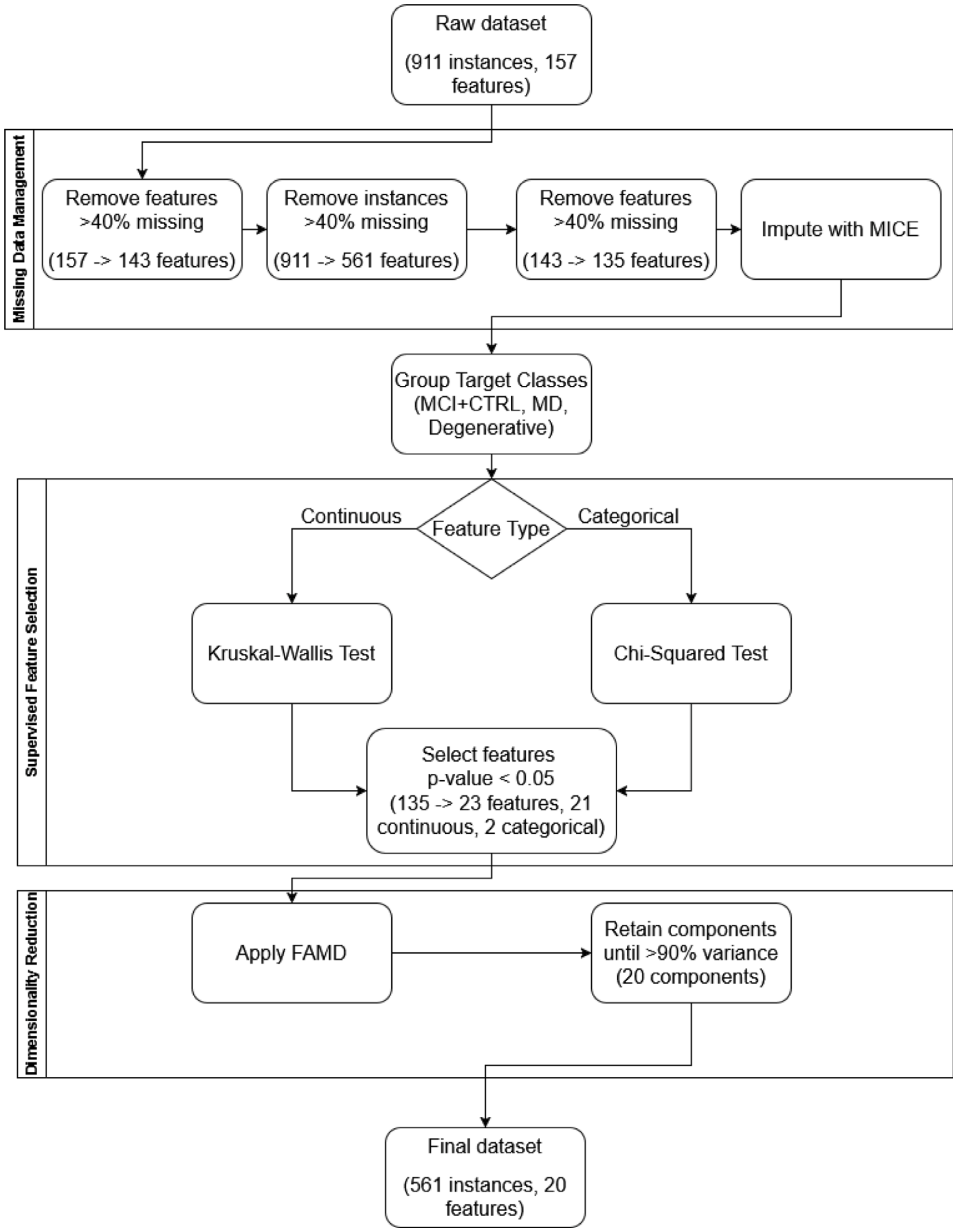

Fig. 1. Flowchart of the pre-processing steps in the data analysis pipeline.

2. Material and methods

In the following sections, we detail the pre-processing and clustering analysis conducted for data analysis. The sequence of all pre-processing operations is illustrated in Fig. 1. The methodology is reported in accordance with the IJMEDI/ChAMAI checklist [25] for reporting medical machine learning studies.

2.1. Dataset analysis

Our analysis was based on a dataset collected at the Policlinico Hospital of Milano (Milan, Italy), between March 2009 and March 2018. The dataset consists of medical records corresponding to 911 anonymized individuals, each described by 157 features. These features are multi-modal, encompassing sociodemographic information, data on the presence or absence of diseases (with a particular focus on the primary target, "Type of Dementia"), clinical test results and signs (excluding blood-related measures), as well as blood test results. Additionally, the dataset includes disability scores, gene expression levels from peripheral blood mononuclear cells, and protein expression levels detected in the patients' blood. The characteristics of the dataset, including key distribution parameters and the number of missing values, are presented in Tables A.7 and A.8, for continuous and categorical features, respectively.

2.2. Missing data management

The dataset contained a substantial number of missing values, as summarized in Tables A.8 and A.7, and visualized in Figures A.3 and A.4. To manage the missing data, we first removed instances and features with a significant amount of missing values to minimize the bias introduced by imputation in subsequent analysis steps.

Initially, we removed columns with more than 40% missing data. This threshold was determined and validated in a previous study [14] based on a reduced version of the dataset (not including the gene and protein expression levels), and was used in this study after consultation with the clinical authors in order to avoid including features with large percentages of missing values that could bias the results of the analysis. This reduced the number of features from 157 to 143. Next, based on the analysis of Figure A.4, we identified two distinct groups of instances, corresponding to the two peaks in the distribution of missing values. The second peak, in particular, was associated with patients for whom genomic and proteomic data were not available. Since the analysis of these data in relation to the target of interest was a primary objective of the study, we selected only those instances with less than 40% missing feature values, so as to exclude patients with significant gaps in genomic and proteomic profile completeness. This reduced the instances from 911 to 561.

Following this selection, 561 instances (out of the initial 911) were retained for further analysis. We also removed features with missing rate greater than 40%, reducing the number of features from 143 to 135. Finally, the remaining missing values were imputed using the MICE (Multivariate Imputation by Chained Equations) algorithm [26], as implemented in the *miceforest* library [27]. The algorithm was run until convergence, to ensure consistency of the results, for a total of two iterations.

To ensure the robustness of the analysis and minimize the potential impact of imputation on subsequent results, we examined the distributions of features before and after imputation. Significant differences in these distributions would indicate that imputation may influence the subsequent analysis, whereas the absence of significant differences would suggest no evidence of such an impact. For this purpose, we employed the multivariate distribution testing algorithm proposed in [28]. We avoided conducting separate distribution tests for each feature to reduce the risk of false rejection of the alternative hypothesis of significant differences due to multiple testing corrections, as outlined in [29].

2.3. Supervised feature selection

As a first step after missing data management, we grouped the classes in the target feature, as some classes were rarely represented. Specifically, following the indications of the clinical authors involved and also based on the medical literature [30,31], we created three class groups: MD (comprising patients diagnosed with mixed dementia, that thus exhibited characteristics of both neurodegenerative and vascular dementia type), MCI+CTRL (comprising patients in the control group or diagnosed with mild cognitive impairment, which either did not have any cognitive deterioration or were otherwise not significantly impaired in their daily life functions), and Degenerative Dementia Types (encompassing the patients that had a confirmed diagnosis of one of the remaining forms of neurodegenerative dementia, wherein AD was the most prevalent one).

Subsequently, given the large number of features in the dataset, we applied various dimensionality reduction techniques to achieve a better ratio between the number of patients and features, thereby mitigating the curse of dimensionality [32] which can negatively influence the results of analyses based on geometric distances (including unsupervised dimensionality reduction [33] and clustering [34]).

As a first dimensionality reduction step, we conducted supervised feature selection based on classical, univariate statistical tests to identify features significantly associated with the target variable. For continuous and ordinal categorical features, we applied the Kruskal–Wallis test [35], and for the remaining categorical features, we used the chi-square test. We selected for further analysis only the features that were significantly associated with the target variable, as determined by a p-value below 0.05 (corresponding to a 95% confidence level) after multiple test correction. Multiple test correction was performed using the Benjamini–Hochberg procedure for false discovery rate correction.

2.4. Unsupervised dimensionality reduction

After conducting supervised feature selection, we also applied unsupervised dimensionality reduction techniques. Since the primary objective of this study was to identify factors influencing the presence of mixed dementia in patients, we chose to use linear feature extraction methods for their superior interpretability. However, due to the multi-modal nature of the dataset – comprising both continuous and categorical – Principal Component Analysis (PCA) [36] was not suitable as a dimensionality reduction technique for analyzing the whole dataset.

To effectively model these multi-modal features, we employed the Factor Analysis of Mixed Data (FAMD) [37] algorithm. FAMD integrates the results of PCA for continuous features with those from Multiple Correspondence Analysis (MCA) [38] for categorical features. This technique was implemented using the *prince* Python library [39]. The key hyperparameter, the number of components, was selected to account for at least 90% of the variance [40] in the data, so as to preserve as much as possible the multivariate relationships in the data.

2.5. Clustering

Following data processing, we conducted an unsupervised analysis of the dataset using clustering algorithms. We explored three different clustering techniques, namely: k-means and spectral clustering [41] (representing partitional clustering methods), and HDBSCAN [42] (a representative density-based clustering method). The primary objective of this unsupervised analysis was to evaluate the potential correspondence between the data-driven categorization of patients (as determined by the clustering algorithms) and the classification target in the analyzed dataset. In particular, we considered the same grouping of classes as previously described: MCI + CTRL, MD, and Degenerative Dementia Types.

For each algorithm we considered different hyperparameter settings. For k-means, we optimized the number of clusters (ranging from 3 to 8). For HDBSCAN, we optimized the minimum cluster size (ranging from 1 to 50), the distance metric (choosing between Euclidean, Manhattan, and Canberra [43,44]), and the distance scaling parameter (ranging from 0 to 2). For spectral clustering, we optimized the number of clusters (ranging from 3 to 8), the affinity function (between nearest neighbors and radial basis function), and the number of neighbors (ranging from 5 to 50). To optimize these parameters, we employed a random search algorithm, conducting 10,000 trials for each clustering algorithm. The optimal clustering algorithm and its hyperparameters were selected based on an internal evaluation procedure using the Davies–Bouldin index [45], where the algorithm and hyperparameter configuration that yielded the lowest Davies–Bouldin score was selected for further analysis.

The best-performing clustering algorithm, with its optimized hyperparameters, was compared with the grouped target variable using the Adjusted Rand Index metric [46]. Additionally, we analyzed the distribution of features across the resulting clusters to assess the presence of statistically significant differences. For continuous and ordinal variables, we applied the Kruskal–Wallis test, while for categorical variables, we used the chi-squared test. P-values were adjusted for multiple hypothesis testing using the Benjamini–Hochberg procedure [47] for false discovery rate control. Significance was evaluated at the 95% confidence level, which means that a feature was considered to be significantly associated with the clustering if the corresponding adjusted p-value was below 0.05. Effect sizes were calculated using the rank-biserial correlation [48] for continuous and ordinal features, and Cohen's D [49] for categorical features.

3. Results

The descriptive statistics for the continuous and categorical features are presented in Table 2, while the distribution of missing data across

Table 2
Results of the supervised feature selection for the features. For continuous variables, results are based on the Kruskal–Wallis test, reporting the test statistic (H), the *p*-value, and the effect size (RBC). For categorical variables, results are based on the Chi-Squared test, reporting the test statistic (chi2), the *p*-value, and the effect size (Cohen D). The feature type (gene expression, proteomic, or other) is also indicated. P-values were adjusted for multiple hypothesis testing using the Benjamini–Hochberg procedure. The full list of features is in Tables A.9 and A.10.

Feature	Statistic (H or chi2)	p	Effect size (RBC or Cohen D)	Nature	Variable type
MMSE	243.947	<0.001	0.524	other	Continuous
IADL total	133.296	<0.001	0.39	other	Continuous
Clock Drawing Test	124.572	<0.001	0.397	other	Continuous
ADL total	52.972	<0.001	0.238	other	Continuous
NfL (pg/ml)	35.158	<0.001	0.214	proteomic	Continuous
SPHK1	34.989	<0.001	0.217	gene expression	Continuous
TNF	30.034	<0.001	0.191	gene expression	Continuous
Education	27.037	<0.001	0.178	other	Continuous
VAMP2	21.869	<0.001	0.204	gene expression	Continuous
IL1B	17.018	0.002	0.136	gene expression	Continuous
UGCG	16.52	0.002	0.177	gene expression	Continuous
ADORA2A	16.67	0.002	0.144	gene expression	Continuous
Age	16.369	0.002	0.163	other	Continuous
IL10 (pg/ml)	15.308	0.003	0.13	proteomic	Continuous
K	14.169	0.005	0.145	other	Continuous
Vitamin D	14.028	0.005	0.142	other	Continuous
ALT GPT	11.757	0.016	0.12	other	Continuous
IL6 (pg/ml)	11.685	0.016	0.129	proteomic	Continuous
Na	10.214	0.031	0.128	other	Continuous
IFNG	10.153	0.031	0.161	gene expression	Continuous
IL6	9.592	0.039	0.113	gene expression	Continuous
Cognitive impairment	324.927	<0.001	1.609	other	Categorical
Gait disorder	27.799	<0.001	0.35	other	Categorical

instances and features is shown in Figures A.3 and A.4. The multivariate hypothesis test applied after missing data imputation returned a *p*-value of 0.867: therefore, there was no evidence of significant differences between the distributions of the features before and after imputation.

The results of the supervised feature selection are reported in Table 2, and the outcomes of the unsupervised dimensionality reduction are provided in Table A.11. Figure A.5 illustrates the results of the dimensionality reduction process, showing moderate class separability. Specifically, two moderately separable subgroups were identified: one comprising the MCI and CTRL classes, and the other including the MD and Degenerative Dementia Types classes.

The results of model selection and hyperparameter optimization, based on the Davies–Bouldin score, as well as the results of the comparison with the ground truth classification, based on the Adjusted Rand Index score, are summarized in Table 3. The best-performing clustering algorithm was k-means, with the optimal number of clusters set to 4. When compared to the ground truth classification (in terms of the three classes: MCI+CTRL, MD, and Degenerative Dementia Types), the clustering produced by k-means showed moderate alignment, with an Adjusted Rand Index of 0.357. The resulting clustering is visualized in Fig. 2, and the distribution of ground truth classes within each cluster is detailed in Table 4. Both clusters 1 and 2 were primarily associated with the MCI+CTRL class. In contrast, clusters 0 and 3 exhibited overlap between the MD and Degenerative Dementia Types classes. The proportions in cluster 0 were significantly different (likelihood ratio test for binomial distributions, *p*-value < 0.001), making it predominantly associated with the MD class, while no significant difference was found in cluster 3 (likelihood ratio test for binomial distributions, *p*-value = 0.377).

In Tables 5 and 6, we present the results of the statistical comparison for the selected features within the clusters identified by the k-means algorithm, for continuous and categorical features, respectively.

We observed significant differences in several proteomic markers' concentrations (IL6, NfL, TNFR1, IL10, TNFa, TREM1, TREM2) and gene expression levels (SPHK1, TNF, ADORA2 A, SMPD4, IL1B, UGCG, IL6, TYROBP, IFNG, VAMP2, NFKB1, NLRP3, DEGS1). Additionally, significant differences were found in various laboratory parameters (K, TSH, Vitamin D, ALT GPT, Na, Folates, AST GOT), as well as clinical variables (MMSE, IADL total, ADL total, Clock Drawing Test, Cognitive deterioration, Gait disorder, Sleep disorders, Cerebrovascular disease,

Table 3
Results of the clustering algorithms, in terms of the Davies–Bouldin score and Adjusted Rand Index, obtained through the hyperparameter selection process and comparison with the grouped ground truth labels. Davies–Bouldin: Lower values are better (i.e., associated with a better clusterization); Adjusted Rand Index: Higher values are better (i.e., better association with the grouped ground truth labels).

Algorithm	Davies–Bouldin score	Adjusted rand index
k-means	2.714	0.357
spectral clustering	2.832	0.229
HDBSCAN	3.964	0.004

Table 4
Percentages of classes within the clusters related to the Fig. 2.

Cluster	MCI+CTRL	MD	Degenerative dementia types	Cluster size
0	10%	62%	28%	86
1	92%	4%	4%	240
2	80%	12%	8%	85
3	18%	45%	37%	150

Anxiety/Depression, Osteoarthritis, Peripheral edema, Pain, Heart failure, Renal insufficiency) and demographic features (Education, Age). Among these, MMSE, IADL total, ADL total, Cognitive deterioration, and Gait disorder were associated with large effect sizes; Clock Drawing Test and Sleep disorders showed medium effect sizes; while the other significant features were associated with small effect sizes.

Regarding the continuous features, cluster 0 (primarily associated with mixed dementia patients) was characterized by the lowest mean values in MMSE, IADL total, ADL total, Clock Drawing Test, SPHK1, TNF, K, ADORA2 A, SMPD4, IL1B, Vitamin D, UGCG, IL6 (gene expression), NFKB1, NLRP3, and AST GOT. In all of these cases, except for ADL total, cluster 3 (mainly associated with dementia patients) showed the second lowest values. Similarly, cluster 0 was characterized by the highest mean values for IL6 (protein concentration), NfL, TNFR1, IL10 (protein concentration), TNFa, TREM1 (protein concentration), TSH, Na, and TREM2. In all of these cases, except for IL10 (protein concentration) and Na, cluster 2 (mainly associated with MCI+CTRL patients) exhibited the second highest values. In contrast, cluster 2 had the highest mean values for Age, TYROBP, IFNG, ALT GPT, VAMP2, and DEGS1. Cluster 0 had the second highest value for Age and TYROBP,

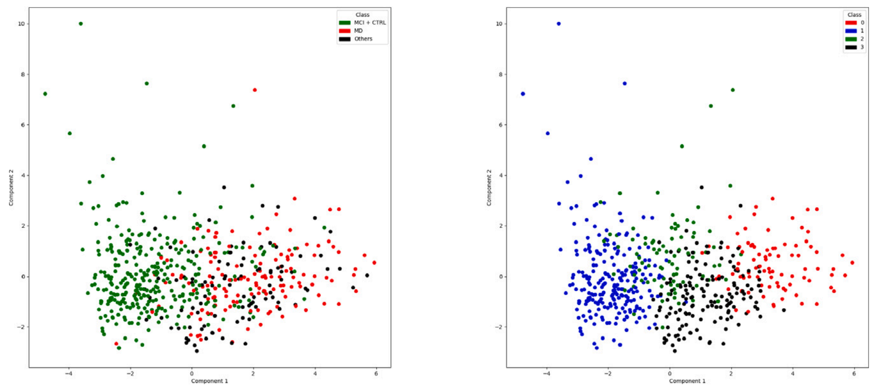

Fig. 2. One the left, assignment of the instances to the grouped class labels. On the right, results of the clustering (k-Means with k=4, seed=642374821). Each instance is represented as a point in the 2D plane determined by the 2 most significant features constructed through the FAMD algorithm. Each instance color represents the corresponding class or cluster assignment.

while cluster 1 (also mainly associated with MCI+CTRL patients) had the second highest values for the remaining features.

For categorical features, cluster 0 had the highest proportion of patients with Sleep disorders (44%, equivalent to cluster 2), Peripheral edema (32%, followed by cluster 3 with 30%), Heart failure (7%), and Chronic renal failure (18%). Additionally, it had the second highest proportion of patients with Cerebral vasculopathy (73%, after cluster 2 at 74%), Cognitive deterioration (92%, after cluster 3 at 97%), Gait disorder (84%, after cluster 2 at 92%), and Pain (27%, after cluster 2 at 38%).

4. Discussion

This study presents a novel application of unsupervised machine learning techniques to analyze a large, multidimensional, and multimodal dataset of dementia patients, with the goal of identifying clinically relevant subtypes within this heterogeneous population. Traditional diagnostic approaches often treat dementia as a monolithic condition. By leveraging clustering methods, our research challenges this perspective and offers new insights into the complexity of dementia and its progression.

The primary objective of this study was to uncover potentially hidden subtypes of dementia by analyzing a dataset of 911 individuals, each characterized by 157 clinical and sociodemographic features. Our analysis identified four distinct clusters within the population, each defined by specific peripheral gene expression, cognitive scores, disability levels, and other clinical factors, thereby providing a more granular understanding of the disease.

One of the most significant findings was the clear differentiation between mild cognitive impairment (MCI) and more severe forms of dementia. The clusters demonstrated a distinct stratification between patients with early-stage dementia and those with advanced cognitive decline, reinforcing the notion that early- and late-stage dementia may represent distinct clinical entities. Notably, the clusters associated with late-stage dementia (clusters 0 and 3) were characterized not only by lower scores on common diagnostic tests (such as MMSE [50], IADL [51], ADL [52], and the Clock Drawing Test [53]), but also by markedly distinct gene expression level and protein blood concentrations, including significantly reduced expression of SPHK1, ADORA2 A, and SMPD4. While some of these biomarkers have been previously studied and linked to disease progression in dementia [54–58], the main innovation in our research lies in the multi-dimensional and multi-modal approach. By integrating clinical, genomic, and proteomic information through clustering analysis, our study provides a comprehensive profiling that differentiates between mild and late-stage forms of the disease. Interestingly, patients in the late-stage dementia clusters also exhibited a pronounced Vitamin D deficiency, a finding corroborated by previous studies [59]. This suggests the potential importance of maintaining normal Vitamin D levels to prevent the onset of dementia.

Another significant finding is the identification of two distinct subtypes within the dementia population, confirming that dementia is more heterogeneous than traditionally thought. These clusters exhibited substantial differences in genomic and proteomic profiles, patient demographics, and clinical characteristics. Although we focused on distinguishing mixed dementia from neurodegenerative dementia types, our clustering analysis revealed that mixed dementia shares overlapping characteristics with both late-stage dementia and mild cognitive impairment. This finding aligns with the understanding that mixed dementia is an overlapping condition with features of both vascular and neurodegenerative diseases [60,61]. In addition to the previously mentioned markers, for which patients in cluster 0 exhibited more extreme profiles than those in cluster 3, cluster 0 also showed increased expression of several other significant biomarkers. These included NfL (studied as a biomarker for neurodegeneration [62]), TNFR1 (previously linked to the progression from mild cognitive impairment to Alzheimer's disease [63]), IL10 (protein concentration) (associated with a protective role against the onset of Alzheimer's [64]), as well as TREM1 (protein concentration) and TREM2 (both proposed as risk factors for Alzheimer's disease [65,66]).

These findings have significant implications for both clinical practice and research. First, by revealing the heterogeneity within dementia populations, this study underscores the limitations of current diagnostic frameworks that treat dementia as a uniform condition. The clusters identified here provide a foundation for revisiting and refining these diagnostic criteria to better account for the variability observed in disease progression and patient outcomes. Notably, the identification of a mixed dementia subtype, characterized by both neurodegenerative and vascular pathologies [4], aligns with recent literature suggesting that vascular contributions to cognitive decline are widespread and often precede the classical neurodegenerative signs of diseases like Alzheimer's. The identification of specific clinical and demographic profiles for each cluster also holds potential for personalizing treatment strategies. Patients in different clusters may respond differently to therapeutic interventions, and care plans tailored to cluster membership could enhance both treatment efficacy and quality of life. This approach could reduce the trial-and-error nature of dementia care, leading to more effective and individualized patient management.

While the primary goal of this study was to distinguish between mild and severe dementia stages, the clustering analysis unexpectedly revealed subgroups with mixed clinical presentations. Some patients exhibited characteristics of both MCI and more advanced forms of dementia, suggesting a more fluid continuum between dementia stages than previously understood. The specific clinical profiles observed provide a more detailed picture of how these mixed forms manifest,

Table 5
Results of the between clusters feature distributions' comparisons, for the continuous features, obtained through the Kruskal–Wallis test. For each feature we report the test statistic, *p*-value, effect size and type (gene expression, proteomic, other), as well as the mean and standard deviation (in parenthesis) for each cluster. P-values were adjusted for multiple hypothesis testing using Benjamini–Hochberg procedure.

Feature	H	p	RBC	Nature	Mean 0	Mean 1	Mean 2	Mean 3
MMSE	340.141	<0.001	0.628	other	19.82 (5.1)	27.76 (1.79)	26.63 (2.42)	20.06 (4.55)
IADL tot	258.667	<0.001	0.633	other	1.65 (1.17)	6.62 (1.57)	5.55 (2.02)	4.54 (2.01)
ADL tot	216.284	<0.001	0.545	other	3.07 (1.45)	5.65 (0.57)	5.07 (0.88)	5.33 (0.78)
Clock Drawing Test	203.987	<0.001	0.451	other	1.47 (1.48)	4.03 (1.31)	3.02 (1.88)	1.57 (1.71)
Education	78.903	<0.001	0.267	other	2.12 (1.05)	2.65 (0.97)	2.0 (0.9)	1.74 (0.99)
IL6 (pg/ml)	39.993	<0.001	0.242	proteomic	8.45 (17.67)	4.35 (5.28)	6.13 (10.14)	5.51 (8.32)
NfL (pg/ml)	38.520	<0.001	0.214	proteomic	59.81 (30.28)	43.19 (27.97)	57.32 (46.08)	48.03 (20.62)
Age	34.940	<0.001	0.208	other	80.85 (5.05)	77.83 (5.51)	80.93 (5.12)	78.83 (5.64)
SPHK1	33.392	<0.001	0.230	gene expression	1.08 (0.69)	1.62 (1.42)	1.65 (1.03)	1.13 (0.7)
TNF	32.958	<0.001	0.201	gene expression	0.43 (0.64)	0.83 (1.02)	0.78 (1.25)	0.5 (0.77)
K	31.836	<0.001	0.223	other	4.2 (0.38)	4.46 (0.38)	4.48 (0.39)	4.4 (0.39)
TNFR1 (pg/ml)	29.896	<0.001	0.211	proteomic	2038.64 (897.02)	1577.45 (576.98)	1774.88 (630.85)	1652.6 (578.4)
IL10 (pg/ml)	29.389	<0.001	0.211	proteomic	3.24 (1.56)	2.71 (1.86)	2.58 (1.51)	2.94 (2.46)
ADORA2A	25.434	<0.001	0.218	gene expression	0.92 (0.44)	1.05 (0.49)	1.38 (0.85)	0.97 (0.43)
TNFa (pg/ml)	19.804	0.001	0.167	proteomic	11.3 (3.38)	9.72 (3.51)	10.42 (3.8)	10.07 (3.26)
SMPD4	19.198	0.001	0.158	gene expression	0.47 (0.26)	0.63 (0.37)	0.56 (0.27)	0.5 (0.28)
TREM1 (pg/ml)	19.267	0.001	0.170	proteomic	629.98 (277.38)	512.88 (177.92)	552.66 (188.46)	519.56 (177.54)
IL1B	19.360	0.001	0.150	gene expression	1.02 (1.22)	3.01 (15.67)	1.99 (2.59)	1.43 (4.26)
TSH	18.729	0.002	0.161	other	3.4 (3.69)	2.23 (1.33)	3.25 (4.44)	2.34 (1.86)
Vitamin D	18.491	0.002	0.120	other	16.47 (10.53)	21.1 (13.29)	19.66 (19.13)	17.58 (15.44)
UGCG	18.099	0.002	0.165	gene expression	0.96 (0.37)	1.1 (0.63)	1.12 (0.41)	0.98 (0.56)
IL6	17.701	0.002	0.148	gene expression	0.78 (0.97)	1.87 (4.83)	1.62 (2.52)	1.02 (2.23)
TYROBP	15.504	0.006	0.148	gene expression	1.82 (0.83)	1.6 (0.73)	2.21 (4.09)	1.59 (0.56)
IFNG	14.688	0.009	0.161	gene expression	0.2 (0.14)	0.21 (0.2)	0.31 (0.32)	0.18 (0.2)
ALT GPT	14.561	0.009	0.127	other	16.78 (9.56)	18.5 (10.17)	18.58 (10.12)	16.57 (10.13)
VAMP2	14.206	0.01	0.149	gene expression	0.71 (0.24)	0.76 (0.25)	0.81 (0.26)	0.71 (0.25)
NFKB1	14.280	0.01	0.162	gene expression	0.69 (0.23)	0.8 (0.3)	0.84 (0.34)	0.75 (0.28)
Na	13.514	0.013	0.161	other	142.52 (2.35)	141.77 (2.56)	141.16 (2.83)	141.67 (2.51)
Folates	13.427	0.013	0.109	other	8.04 (13.0)	8.16 (6.77)	7.49 (4.95)	6.76 (4.91)
NLRP3	13.117	0.015	0.118	gene expression	0.99 (0.95)	1.87 (2.62)	1.41 (1.58)	1.11 (1.53)
AST GOT	12.322	0.02	0.124	other	19.87 (7.25)	21.75 (9.63)	21.35 (8.43)	21.25 (12.25)
TREM2 (pg/ml)	11.513	0.029	0.123	proteomic	47961.49 (18805.96)	40755.92 (15289.28)	45926.01 (21633.43)	42463.99 (13979.69)
DEGS1	11.431	0.029	0.137	gene expression	1.11 (0.32)	1.12 (0.3)	1.19 (0.26)	1.08 (0.27)

potentially offering a means to predict which patients are at higher risk of rapid progression to severe dementia. Interestingly, certain clinical features, such as specific cognitive test scores or functional impairment measures, were strongly predictive of cluster membership. These

Table 6
Results of the between clusters feature distributions' comparisons, for the categorical features, obtained through the chi-squared test. For each feature we report the test statistic, *p*-value, effect size and type (genomic, proteomic, other), as well as the mode and normalized entropy (in parenthesis, as a measure of variation) for each cluster. The normalized entropy was calculated by dividing the Shannon entropy by its maximum value. P-values were adjusted for multiple hypothesis testing using Benjamini–Hochberg procedure.

Feature	chi2	p	Cohen D	Nature	Mode 0	Mode 1	Mode 2	Mode 3
Cognitive deterioration	408.583	<0.001	2.209	other	1 (0.41)	0 (0.37)	0 (0.65)	1 (0.18)
Gait disorder	356.361	<0.001	2.236	other	1 (0.64)	0 (0.12)	1 (0.41)	0 (0.69)
Sleep disorders	27.119	<0.001	0.335	other	0 (0.99)	0 (0.75)	0 (0.99)	0 (0.83)
Cerebral vasculopathy	18.204	0.003	0.261	other	1 (0.84)	1 (1.0)	1 (0.82)	1 (0.96)
Anxiety/Depression	13.674	0.021	0.263	other	1 (1.0)	0 (0.99)	1 (0.97)	0 (0.95)
Osteoarthritis	12.699	0.028	0.242	other	1 (1.0)	0 (0.97)	1 (1.0)	0 (0.93)
Peripheral edema	11.976	0.029	0.189	other	0 (0.9)	0 (0.66)	0 (0.82)	0 (0.88)
Pain	12.163	0.029	0.243	other	0 (0.84)	0 (0.79)	0 (0.96)	0 (0.69)
Heart failure	11.153	0.038	0.200	other	0 (0.37)	0 (0.07)	0 (0.09)	0 (0.18)
Chronic renal failure (IRC)	10.582	0.044	0.183	other	0 (0.69)	0 (0.38)	0 (0.41)	0 (0.38)

results align with other machine learning studies that emphasize the importance of cognitive markers in predicting dementia outcomes [67]. However, the integration of genomic and proteomic data in this study adds a crucial dimension, indicating that biological factors can further refine our ability to predict dementia trajectories more accurately.

Despite the strengths of this study, several limitations must be acknowledged. First, while the unsupervised machine learning approach provided valuable insights, the resulting clusters require further validation in independent cohorts. External validation is necessary to confirm that these subtypes are not artifacts of the specific dataset or clustering technique employed. Additionally, although the dataset included a wide range of clinical, demographic, and biological features, there were still gaps, particularly in the proteomic and genomic data, where some values were imputed rather than directly measured. Future studies should aim to incorporate more comprehensive molecular data to strengthen the robustness of subtype classification.

Moreover, the clustering approach used in this study, while effective in identifying distinct subgroups, could be improved by integrating longitudinal data to capture changes in cluster membership over time. This would allow for a more dynamic understanding of how patients progress through different stages of dementia and whether certain interventions can alter the course of the disease. Additionally, while the unsupervised approach identified significant differences in genomic and proteomic profiles across clusters, it does not address the causal direction of these expression changes. Further research is needed to determine whether these alterations emerge as a consequence of dementia or if they are associated with biological mechanisms driving the onset and progression of the disease. This could pave the way for targeted therapeutic interventions and personalized management of different dementia subtypes.

Additionally, we note that the population considered in this study mainly consists of older patients: while this is expected, as dementia is a disease traditionally associated with older age, at the same time it makes the characterization of dementia subtypes harder since, usually, multiple forms of dementia can coexist in the same patient.

Finally, this study underscores the need for more personalized research in dementia care. As cluster-based approaches become more refined, future research should focus on developing predictive models that can be applied in clinical settings to identify patient subtypes at the point of diagnosis. In parallel, clinical trials should be designed to assess whether treatments tailored to these subtypes result in improved outcomes compared to traditional, one-size-fits-all approaches.

5. Conclusion

This study demonstrates the potential of unsupervised machine learning to uncover clinically relevant subtypes within the dementia population, challenging the traditional, homogeneous classification of the disease. By analyzing a multi-modal, high-dimensional dataset that includes diverse clinical, demographic, genomic, and proteomic features, we identified distinct subgroups with significant implications for diagnosis and personalized care. These findings underscore the importance of shifting toward more individualized approaches to dementia care, which could enhance diagnostic accuracy and optimize treatment strategies. Future research should focus on validating these subtypes in diverse populations and exploring their therapeutic potential in clinical practice. This could ultimately transform dementia care and improve outcomes for patients across all stages of the disease.

CRediT authorship contribution statement

Andrea Campagner: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Luca Marconi:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Edoardo Bianchi:** Writing – original draft, Visualization, Validation, Software, Formal analysis. **Beatrice Arosio:** Writing – review & editing, Validation, Investigation, Data curation, Conceptualization. **Paolo Rossi:** Writing – review & editing, Supervision, Data curation, Conceptualization. **Giorgio Annoni:** Writing – review & editing, Supervision, Data curation, Conceptualization. **Tiziano Angelo Lucchi:** Writing – review & editing, Supervision, Data curation, Conceptualization. **Nicola Montano:** Writing – review & editing, Supervision, Funding acquisition, Data curation, Conceptualization. **Federico Cabitza:** Writing – review & editing, Visualization, Validation, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Acronyms

- **ACE**: Angiotensin-Converting Enzyme
- **ADAM10**: A Disintegrin and Metalloproteinase 10
- **ADL**: Activities of Daily Living
- **ADORA2A**: Adenosine A2 A Receptor
- **ADORA2B**: Adenosine A2B Receptor
- **ALT**: Alanine Aminotransferase
- **ApoE**: Apolipoprotein E
- **AST**: Aspartate Aminotransferase
- **BDNF**: Brain-Derived Neurotrophic Factor
- **BMI**: Body Mass Index
- **CASP1**: Caspase 1

- **CASP8**: Caspase 8
- **CERS2**: Ceramide Synthase 2
- **CKD**: Chronic Kidney Disease
- **CLCN2**: Chloride Voltage-Gated Channel 2
- **CLOCK**: Circadian Locomotor Output Cycles Kaput
- **COL6A1**: Collagen Type VI Alpha 1 Chain
- **COL6A2**: Collagen Type VI Alpha 2 Chain
- **COL6A3**: Collagen Type VI Alpha 3 Chain
- **COPD**: Chronic Obstructive Pulmonary Disease
- **CRP**: C-Reactive Protein
- **DEGS1**: Delta 4-Desaturase Sphingolipid 1
- **EMILIN1**: Elastin Microfibril Interfacer 1
- **EMILIN2**: Elastin Microfibril Interfacer 2
- **FLNA**: Filamin A
- **fMRI**: functional Magnetic Resonange Imaging
- **GDS**: Geriatric Depression Scale
- **HDLBP**: High Density Lipoprotein Binding Protein
- **HIF1A**: Hypoxia-Inducible Factor 1-alpha
- **HIF1AN**: Hypoxia-Inducible Factor 1-alpha Inhibitor
- **IADL**: Instrumental Activities of Daily Living
- **IFNG**: Interferon Gamma
- **IGF1R**: Insulin-like Growth Factor 1 Receptor
- **IL10**: Interleukin-10
- **IL18**: Interleukin-18
- **IL1B**: Interleukin-1 Beta
- **IL1R1**: Interleukin-1 Receptor Type 1
- **IL6**: Interleukin-6
- **IL6R**: Interleukin-6 Receptor
- **K**: Potassium
- **MCH**: Mean Corpuscular Hemoglobin
- **MCV**: Mean Corpuscular Volume
- **MMSE**: Mini-Mental State Examination
- **MRI**: Magnetic Resonance Imaging
- **Na**: Sodium
- **NFKB1**: Nuclear Factor Kappa B 1
- **NfL**: Neurofilament Light Chain
- **NLRP3**: NOD-like Receptor Family Pyrin Domain Containing 3
- **NOS3**: Nitric Oxide Synthase 3
- **PIN1**: Peptidylprolyl Cis/Trans Isomerase NIMA-Interacting 1
- **PLD3**: Phospholipase D Family Member 3
- **SERPINI1**: Serpin Family I Member 1
- **SGMS1**: Sphingomyelin Synthase 1
- **SIGIRR**: Single Immunoglobulin Interleukin-1-Related Receptor
- **SMPD1**: Sphingomyelin Phosphodiesterase 1
- **SMPD3**: Sphingomyelin Phosphodiesterase 3
- **SMPD4**: Sphingomyelin Phosphodiesterase 4
- **SPARC**: Secreted Protein Acidic And Cysteine Rich
- **SPHK1**: Sphingosine Kinase 1
- **SPTLC1**: Serine Palmitoyltransferase Long Chain Base Subunit 1
- **SPTLC2**: Serine Palmitoyltransferase Long Chain Base Subunit 2
- **SPTLC3**: Serine Palmitoyltransferase Long Chain Base Subunit 3
- **ST3GAL5**: ST3 Beta-Galactoside Alpha-2,3-Sialyltransferase 5
- **STX1A**: Syntaxin 1A
- **STX3**: Syntaxin 3
- **SYK**: Spleen Associated Tyrosine Kinase
- **TGFA**: Transforming Growth Factor Alpha
- **TGFB1**: Transforming Growth Factor Beta 1
- **TGFB2**: Transforming Growth Factor Beta 2
- **TIMP1**: Tissue Inhibitor of Metalloproteinase 1
- **TNF**: Tumor Necrosis Factor
- **TNFR1**: Tumor Necrosis Factor Receptor 1
- **TP53**: Tumor Protein p53
- **TP53BP2**: Tumor Protein p53 Binding Protein 2
- **TP73**: Tumor Protein p73
- **TREM1**: Triggering Receptor Expressed on Myeloid Cells 1
- **TREM2**: Triggering Receptor Expressed on Myeloid Cells 2
- **TREML1**: Triggering Receptor Expressed On Myeloid Cells Like 1
- **TREML2**: Triggering Receptor Expressed On Myeloid Cells Like 2
- **TSH**: Thyroid-Stimulating Hormone
- **TYROBP**: TYRO Protein Tyrosine Kinase Binding Protein
- **UGCG**: UDP-Glucose Ceramide Glucosyltransferase
- **VAMP2**: Vesicle-Associated Membrane Protein 2
- **VCAM1**: Vascular Cell Adhesion Molecule 1
- **VDR**: Vitamin D Receptor
- **VEGFA**: Vascular Endothelial Growth Factor A
- **VEGFB**: Vascular Endothelial Growth Factor B
- **VEGFC**: Vascular Endothelial Growth Factor C

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.jbi.2025.104799.

Data availability

Code and data supporting the analysis are available upon reasonable request to the authors.

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